



KAMBALA

Mathematics Extension 2

HSC Assessment Task 1

November 2006

Time Allowed: 1 hour

Outcomes Assessed

- E2** chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings
- E3** uses the relationship between algebraic and geometric representations of complex numbers
- E6** combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions
- E9** communicates abstract ideas and relationships using appropriate notation and logical argument

INSTRUCTIONS

- This task contains 3 questions of 12 marks each. Marks for each part question are shown.
- Answer all questions on the writing paper provided. **Start each question on a new page.**
- Calculators may be used.
- Show all necessary working.
- Marks may be deducted for careless or badly arranged work.

Student ID: _____

Question 1 (Start a new page.)

12 Marks

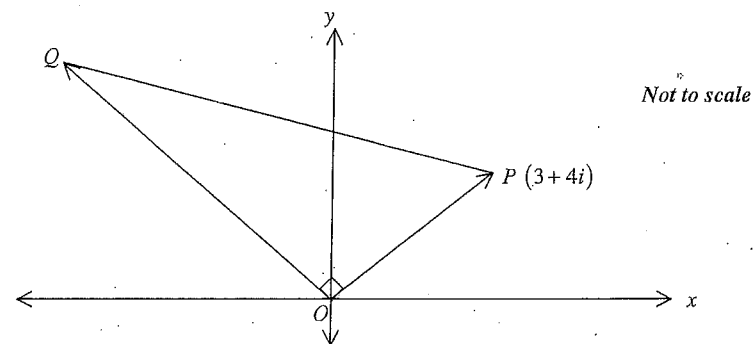
- (a) For the complex number $z = 2 - i$, express the following in the form $x + iy$ where x and y are real:

(i) $\bar{z}$ 1

(ii) $\frac{1}{z}$ 1

(iii) $(1 - i)z$ 2

(b)



The triangle POQ above is right-angled at O . The length of OQ is twice that of OP . Given that $\overline{OP}$ represents the complex number $3 + 4i$, determine the complex number represented by $\overline{OQ}$. 2

- (c) Find u and v , where u and v are real, if $(u - iv)^2 = 5 + 12i$. 3

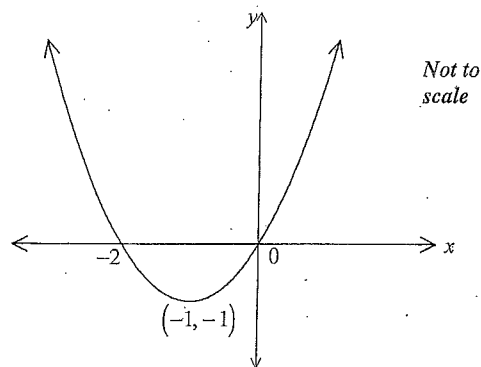
- (d) By factorising $z^3 + 8$, or otherwise, find all solutions of $z^3 + 8 = 0$. Express any complex roots in the form $r(\cos \theta + i \sin \theta)$ where r is real and θ is in radians. 3

Question 2

(Start a new page.)

12 Marks

- (a) Let $f(x) = x(x+2)$. The graph of $y = f(x)$ is drawn below.



On separate diagrams, and without using calculus, sketch the following graphs. Indicate clearly any asymptotes and intercepts with the axes.

(i) $y = f(x) + 2$

1

(ii) $y = |f(x)|$

2

(iii) $y = \frac{1}{f(x)}$

2

(b) Let $f(x) = \frac{x^2}{(x+2)(x-2)}$ for $x \neq 0$

(i) Show that $f(x) = 1 + \frac{4}{x^2 - 4}$

1

(ii) Find the equations of all asymptotes to $y = f(x)$.

2

(iii) Using part (i) find any stationary points on the curve and determine their nature.

2

(iv) Sketch the curve $y = f(x)$. Clearly indicate any asymptotes and intercepts.

2

Question 3

(Start a new page.)

12 Marks

(a) Write $z = 5\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$ in Cartesian form.

1

(b) Given z is a complex number such that $5z + \bar{z} = 12 + 8i$, find z .
Let $z = x + iy$ where x and y are real.

2

(c) (i) Sketch $|z - 2| = 1$, and describe the locus geometrically.

2

(ii) Considering z as a point on the locus indicate on your sketch the points that give the greatest values of $|z|$ and of $\arg(z)$. Hence find the greatest value of $|z|$ and $\arg(z)$.

2

(d) (i) On the same diagram sketch the graphs of $y = |x + 2|$ and $y = x^2$.

1

(ii) Show that the graphs intersect at $x = -1$ and $x = 2$.
Hence solve $|x + 2| > x^2$ for all x .

2

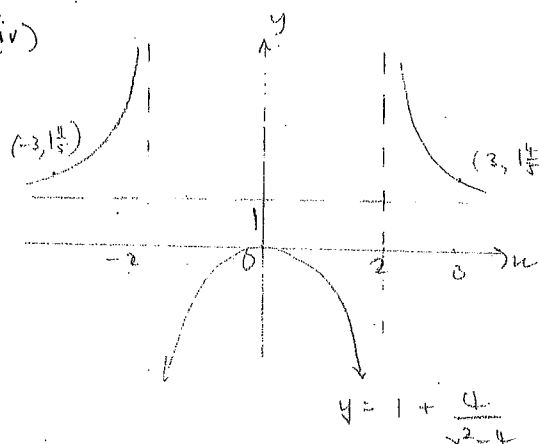
(iii) For what values of b will $|x + 2| = x^2 + b$ have no solution?

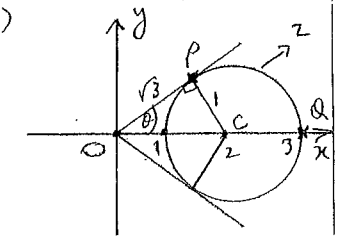
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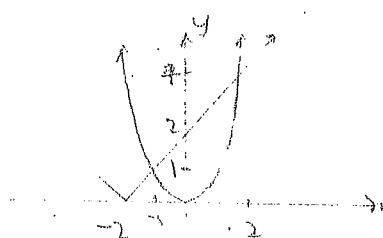
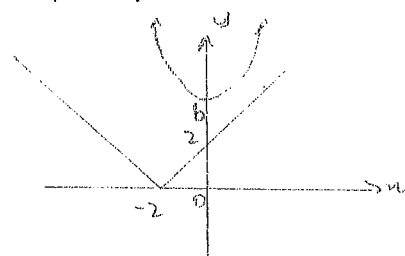
End of Assessment Task

Question	Solutions	Marks	Marking Criteria
1.(a)	$z = 2 - i$ (i) $\bar{z} = 2 + i$ (ii) $\frac{1}{z} = \frac{1}{2-i} \times \frac{2+i}{2+i}$ $= \frac{2+i}{4+1}$ $= \frac{2+i}{5}$ (iii) $(1-i)z$ $= (1-i)(2-i)$ $= 2 - 3i - 1$ $= 1 - 3i$		
(b)	$\vec{OQ} = 2 \cos \frac{\pi}{2} \times \vec{OP}$ $= 2i \times (3+4i)$ $= 6i + 8i^2$ $= -8 + 6i$		
(c)	$(u-iv)^2 = 5+12i$ $(u^2-v^2) - (2uv)i = 5+12i$ $\left. \begin{array}{l} u^2-v^2 = 5 \\ 2uv = -12 \end{array} \right\}$ $uv = -6$ $v = -\frac{6}{u}$ $u^2 + \frac{36}{u^2} = 5$		

Question	Solutions	Marks	Marking Criteria
1(c) c/d	$u^4 - 5u^2 + 36 = 0$ $(u^2 - 9)(u^2 + 4) = 0$ $u^2 = 9 \quad u^2 = -4$ $u = \pm 3 \quad \text{no soln.}$ $u = 3, v = -2$ $u = -3, v = 2$		
(d)	$z^3 = -8$ $z^3 = 8 \cos \pi$ $= 8 \cos(\pi + 2k\pi)$ $z = 2 \cos \frac{(2k+1)\pi}{3}$ where k is an integer $k=0: z = 2 \cos \frac{\pi}{3}$ $k=1: z = 2 \cos \pi = -2$ $k=2: z = 2 \cos \frac{5\pi}{3}$ <u>OR</u> $z^3 + 8 = (z+2)(z^2 - 2z + 4)$ $z = -2, \frac{2 \pm \sqrt{-12}}{2}$ $z = -2, 1 \pm \sqrt{3}i$ $z = 2 \cos \frac{\pi}{3}, 2 \cos \pi$ $2 \cos \frac{5\pi}{3}$		

Q. No.	Solutions	Marks	Comments+Criteria								
2	ad (0,0): <table border="1" data-bbox="257 79 649 223"> <tr> <td>x</td><td>-1</td><td>0</td><td>1</td></tr> <tr> <td>$P'(x)$</td><td>$\frac{8}{9}$</td><td>0</td><td>$-\frac{8}{9}$</td></tr> </table> / - \ $\therefore$ max. at (0,0) (iv)  $y = 1 + \frac{4}{x^2 - 4}$ $f(x) = 1 + \frac{4}{(x)^2 - 4}$ $= 1 + \frac{4}{x^2 - 4}$ $= f(x)$ even	x	-1	0	1	$P'(x)$	$\frac{8}{9}$	0	$-\frac{8}{9}$	1	
x	-1	0	1								
$P'(x)$	$\frac{8}{9}$	0	$-\frac{8}{9}$								
		2									

Question	Solutions	Marks	Marking Criteria
3(a)	$z = 5 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$ $= 5 \left(-\frac{\sqrt{3}}{2} + i \cdot \frac{1}{2} \right)$ $z = -\frac{5\sqrt{3}}{2} + \frac{5}{2}i$		
(b)	$5z + \bar{z} = 12 + 8i$ $5(x+iy) + (x-iy) = 12 + 8i$ $5x + 5iy + x - iy = 12 + 8i$ $6x + i(4y) = 12 + 8i$ $6x = 12 \quad 4y = 8$ $x = 2 \quad y = 2$ $\therefore z = 2 + 2i$		
(c)	(i)  locus is a circle centre (2,0) radius 1 (ii) $ z _{\max} = \vec{OQ} $ $= \sqrt{3}$ $\arg z_{\max} = \theta$ $= \tan^{-1}(1/\sqrt{3})$ $= \pi/6$		

Qn	Solutions	Marks	Comments+Criteria
3 ad	(i) $y = x+2$ and $y = x^2$  (ii) $x^2 = x + 2$ $x^2 - x - 2 = 0$ $(x-2)(x+1) = 0$ $x = 2, -1$ $ x+2 > x^2$ $-1 < x < 2$ (iii) $x+2 > x^2 + b$ have no solns  $x + 2 = x^2 + b$ $x^2 - x + b - 2 = 0$ $\Delta = b^2 - 4ac$ $= 1 - 4 \cdot 1 \cdot (b-2)$ $= 1 - 4b + 8$ $= 9 - 4b$ For no solns $\Delta < 0$ i.e. $9 - 4b < 0$		

$$-4b < -9$$

$$4b > 9$$

$$b > \frac{9}{4} = 2\frac{1}{4}$$

$$\text{i.e. } b > 2\frac{1}{4}$$